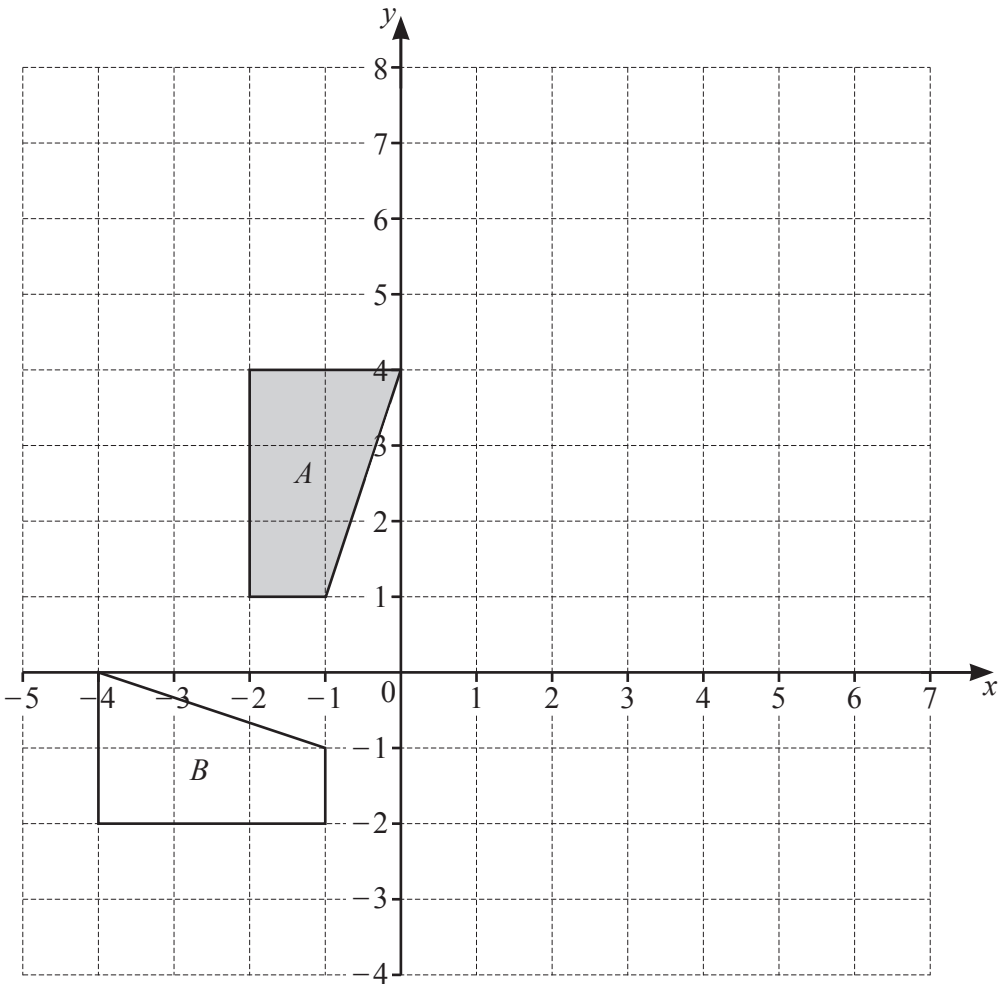
0580_w21_qp_41_Q7

7 **(a)**



(i) On the grid, draw the image of

(a) shape A after an enlargement, scale factor 2, centre (0, 1), [2]

(b) shape A after a reflection in the line $y = x - 1$. [3]

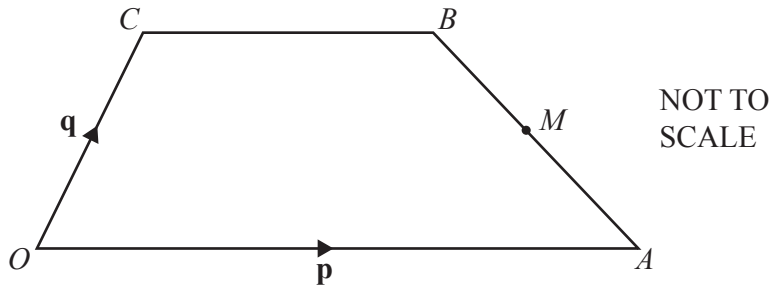
(ii) Describe fully the **single** transformation that maps shape A onto shape B.

.....

..... [3]

0580_w21_qp_41_Q7

(b)



$OABC$ is a trapezium and O is the origin.
M is the midpoint of AB.
$\overrightarrow{OA} = \mathbf{p}$, $\overrightarrow{OC} = \mathbf{q}$ and $OA = 2CB$.

Find, in terms of **p** and **q**, the position vector of M.
Give your answer in its simplest form.

..... [3]